

Production Control Policies to maintain Service Levels in Different Seasons

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Abstract

We analyze a manufacturing system that produces a single type of product to serve multiple classes of customers with seasonal demands. The manufacturer has the flexibility to vary production rates to adapt to seasonal demands. We model the seasonal demand using a Markov modulated Poisson process, and analyze the underlying Markov decision problem to derive optimal production and inventory control policies for the manufacturer. We show that the optimal policy is characterized by a season dependent inventory threshold vector for adjusting production rates and rationing inventory. Further, we impose service level constraints according to the customer classes in each season, and analyze the impact of the service level constraints on the optimal policy and cost under different seasonal demand conditions.

Keywords: Production control, seasonal demands, service levels, dynamic programming

1 Introduction

Our research is motivated by collaborations with several small and medium sized manufacturers (SMMs) that form Tier 1 suppliers to manufacturers such as John Deere, Harley-Davidson, Caterpillar, etc. These SMMs often experience seasonal demands due to various factors such as market conditions, sales initiatives, seasonal behavior of customers, or model changes. In many cases, these demand variations could be significant: the demand during a peak season could be twice the demand observed in a low season. In addition to the challenges posed by such seasonal variations, the suppliers also have to deal with multiple classes of customers that have different utility or value for a given product. These are often

reflected through expectations in delivery lead time and/or penalties for stockouts. These expectations put further stress on manufacturing operations that have to strike a balance between internal production efficiencies and external constraints imposed due to seasonality and customer expectations. Manufacturers usually respond to these challenges by investing in capacity flexibility. We analyze such settings to understand how manufacturers could leverage flexible capacities to deal with challenges posed by seasonality in demands and varying service level expectations.

Manufacturers face high variation in demands across the seasons since the demand at any given time period is the superposition of demands of multiple classes of customers. For instance, SMM supplies the same component to an assembly facility that processes orders for new equipment as well as to dealers that place orders for spare parts to repair old equipment. Orders for spare parts are low in volume but come with a high expectation in terms of delivery and stringent penalties for delays or stock outs. In contrast, orders from the assembly facility are higher in volume and total revenue, and hence deserve a high degree of importance. Further, the demands of each class of customer, in turn, have their own seasonal characteristics. For instance, the demands for a particular component (transmission, gear) from a facility assembling new farm equipment and demands for the same component from a dealer repairing used equipment might have seasons that are synchronized so that the peak demand seasons overlap (over summer). Alternatively, if the same component is used in both lawn mowers and snow blowers, the demands might have complementary seasonality balancing each other.

Further, both the assembly facility and the spare parts dealer might impose contractual conditions that guarantee certain desired service levels and impose penalties for stockouts. When overall workloads on the SMM fluctuates over the year, the SMM has to determine when to leverage their capacity flexibility and incur the costs of overtime (second shift, weekend shifts, adding additional temporary workforce) and when to rely on inventory flexibility to build ahead during low demand periods and incur the additional costs (inventory holding, warehouse operations, and obsolescence). Even when inventory is available, the manufacturer must decide whether the available inventory is enough to satisfy demands in subsequent seasons or if inventory must be rationed in particular seasons among various customer classes to meet the service level expectations in the peak season.

These challenges could be generalized to the following questions. In a manufacturing setting supporting multiple classes of customers with seasonal demands, how should the capacity flexibility investments be leveraged to optimally respond to seasonal demand trends? How

should a system respond to scenarios where demands of multiple customers have synchronized seasons? To what extent do complementary seasons of customer demands help the manufacturing system lower costs? When customers impose individual service level requirements, how would the optimal policies vary — to what extent should the system build in advance in particular seasons versus rely on leveraging its capacity flexibility? We aim to address these questions in this paper.

The rest of the paper is organized as follows. Section 2 provides literature review on work related to serving multiple classes of customers and seasonal demands. Section 3 provides an analytical model for a manufacturing system with flexible capacity to deal with seasonal demands from multiple classes of customers. For this system, we analyze the underlying Markov decision problem and show the existence of the optimal solution and policy under the infinite horizon discounted expected cost and average expected cost criteria. In Section 4, we provide additional properties of the optimal solution and derive an analytical expression for the optimal policy. Though these results are not surprising and may be roughly predicted from literature, our contribution lies in providing a formal proof for these results and addressing all the technical nuances, which have not been shown before for such systems. In Section 5, we determine the impact of service level constraints imposed by the customers on the optimal policy. In Section 6, we derive managerial insights based on the interactions between the seasonal demand patterns and service level constraints of different customer classes. Finally, we provide concluding remarks in Section 7.

2 Relevant Literature

Most of the related literature can be classified into two groups: (i) studies that analyze systems with single class of customers and seasonal demands and (ii) studies that analyze manufacturing systems with multiple classes of customers with stationary demand (no seasonality). For instance, Song and Zipkin (1993) determine optimal ordering policies in a manufacturing system with a single class of customer with seasonal demands and stochastic lead times that are independent of the order quantities. They show that for linear ordering costs, a phase dependent base-stock policy is optimal while, for fixed ordering costs, a phase dependent (s, S) policy is optimal. Under similar settings, but with bounded ordering size and constant lead time, Kapuscinski and Tayur (1998); Yossi and Federgruen (1997) show that a cyclic up-to level policy is the optimal inventory replenishment policy. Hu et al. (2004) analyzes production-inventory system with finite, variable production rates in a seasonal demand setting where demands and production processes are deterministic. They show that

the optimal production policy is a two level inventory hedging policy. Capacity flexibility in the context of service centers has been analyzed by Kumar et al. (2013). They model seasonal demands as a Markov-modulated Poisson process and use dynamic programming to identify the optimal policy as one where capacity is varied based on the system state. Bhat and Krishnamurthy (2015) analyze production systems where capacity and inventory flexibility are leveraged to satisfy seasonal demands of customers. They show that the optimal policy is a season dependent target inventory level policy with state dependent production rates. Bhat and Krishnamurthy (2016) determine different seasonal characteristics and analyze the impact of their joint interaction on a production system, which is a generalized version of the production system analyzed in Bhat and Krishnamurthy (2015). Diaz et al. (2016) analyze a stochastic inventory model with auto-correlated demands and lost-sales feature. The auto-correlated demand is modeled as a Markov-modulated demand, and near-optimal solutions are determined using simulation-based optimization methods.

There are several studies that analyze manufacturing systems with multiple classes of customers with stationary demands (no seasonality). Ha (1997a) analyzes make-to-stock production system with several customer classes and lost sales and shows that the optimal policy is of control limit type with base-stock policy for production decisions and class dependent rationing levels for inventory allocation decisions. Ha (1997b) extends the analysis to multi-class systems that permits backordering of unsatisfied demands. For systems with only two customer classes, they characterize the inventory allocation policy by a single monotone switching curve. Ha (2000) analyzes a make-to-stock production system modeled as an $M/E_k/1$ queue to analyze the impact of low variability in service times. ElHafsi et al. (2010) analyzes a system that is similar to Ha (2000). In addition to modeling the production time as a random variable with Erlang distribution, they also model the inter-arrival time of the customer demands of each class as a random variable with Erlang distribution. De Véricourt et al. (2002) also analyze make-to-stock systems with multi-class customers and backordering of unsatisfied demands. Huang and Iravani (2008) examine the impact of non-unitary order sizes on the manufacturing system's optimal cost and policy under two conditions: i) lost sales environment with multiple customer classes and random batch size, ii) backordering environment with only two customer classes and fixed batch size. Xu et al. (2010) extend the analysis of Huang and Iravani (2008) to incorporate multiple customer classes with backordered demands and varying order sizes arriving demands. They show that the optimal rationing policy is a critical level policy which is a generalized version of the multilevel rationing policy proposed in De Véricourt et al. (2002). However, none of these studies consider systems that have capacity flexibility. Mayorga et al. (2006) analyzes

a system with flexibility in production rates and determines the optimal policy that adjusts capacity and rations inventory in a system with two customer classes, but under stationary demand conditions.

Gayon et al. (2009) analyze a system with multiple customer classes where unmet demands are backordered. They considered two cases depending on whether the production can be interrupted or not. For the uninterrupted production case, they fully characterize the optimal policy based on work storage level, but only partially characterize the optimal policy for the interrupted production case. Benjaafar et al. (2010) analyze a system with two customer classes system where the manufacturer can satisfy, backorder, or reject an arriving customer demand. They characterize the structure of the optimal policy and demonstrate the value in the flexibility to either backorder or reject an arriving customer demand. Zhou et al. (2011) analyze a manufacturing system with finite capacity serving multiple customer classes under lost sales environment. They showed the optimality of a modified based stock policy with multi-level rationing. In contrast to these above work, we analyze systems with endogenous lead time and flexible production capacity under seasonal demand conditions.

Several research also focus on developing heuristic policies that are computationally less intensive for multi-class manufacturing systems (see Fadiloğlu and Bulut (2010), Möllering and Thonemann (2010), Hung and Hsiao (2013), Wang et al. (2013)).

We highlight two papers that are closely related to this paper. Yoon and Lewis (2004) analyze admission control policies for a service system with multiple customers, where the demand process is modeled as a non-homogeneous Poisson process. Customers are classified and admitted according to their reservation price. They identify the optimal policy that determines the admission pricing level based on the time period and queue length. Feng and Pang (2010) analyze production systems with fixed capacity that serve customers with different demand patterns. They characterize the optimal policy as a structured threshold policy for production scheduling and demand fulfillment decisions. In this paper, we focus on manufacturing systems where the decision maker could control both production rates and inventory rationing decisions to serve multiple classes of customers. Additionally, modeling the demand process as MMPP instead of non-homogenous Poisson process, provides us flexibility in capturing different types of seasonal demand patterns.

3 System Description and Problem Formulation

We analyze a system that consists of a manufacturing station and produced goods inventory buffer. The production time at the manufacturing station is exponentially distributed with mean μ^{-1} , implying a production rate of μ . We assume that the manufacturer can vary this rate, μ between 0 and the maximum rate $\mu_{max} < \infty$. The manufacturing system stocks the produced goods in an inventory buffer. Arriving demands are satisfied from available inventory in the buffer. Customer demands that are not satisfied immediately are lost and the manufacturer incurs a corresponding penalty cost for the lost sale. Although the system manufactures a single product, customer demands are classified into l classes of customers based on their penalty cost and p_i represents the penalty cost for class i customer. This penalty cost, p_i reflects the revenue lost by not satisfying a demand for a class i customer, $i = 1, \dots, l$. Without loss of generality, we assume $p_1 > p_2 > \dots > p_l$. This implies that class 1 customers have the highest penalty cost and class l customers have the least penalty cost. In order to minimize the total expected cost, the decision maker continuously reviews the system, adjusts the production rate, if necessary, and determines how to use available inventory to satisfy demands of different classes of customers.

Demands for each customer class are seasonal with distinct average demands in each of the m seasons. We assume that the demands from a customer class are independent of the demands from other customer classes. We model the seasonal demand as a Markov modulated Poisson process (MMPP) with m phases. Each phase in the MMPP corresponds to a distinct arrival rate for each customer class. Demand arrival rate for class i customer in each season s is denoted as $\lambda_{i,s}$. Thus, demand arrivals for class i customers in season $s \in \{1, \dots, m\}$ is a Poisson process with arrival rate $\lambda_{i,s} < \infty$.

Note that a change in the season corresponds to a phase transition in the MMPP. This phase transition process is modeled as a finite state continuous time Markov chain with irreducible generator matrix Q . The state space for this Markov chain is $\mathbb{S} := \{1, 2, \dots, m\}$. We assume that the phase transition process is ergodic and has unique stationary probability distribution vector denoted by $\{\hat{\pi}_1, \hat{\pi}_2, \dots, \hat{\pi}_m\}$.

The dynamics of the system is modeled as a continuous-time Markov chain (CTMC) since the inter-arrival time of each customer class, production time, and the season length are exponentially distributed. Hence, we formulate a Markov decision problem (MDP) to identify the optimal policy for the manufacturing system. In particular, we develop a discrete epoch

Markov decision problem with the underlying CTMC using the uniformization technique described in Lippman (1975). The Markov decision problem is formulated as follows:

Decision epochs: The decision epochs $k = 1, 2, \dots$ correspond to instances of arrivals, production completion and phase transitions in the arrival process.

State space: The state space is modeled as a two dimensional Markov chain $\mathbb{X} = \{(n, s) | n \in \mathbb{Z}^+, s \in \mathbb{S}\}$, where n is the inventory of the system and s is the demand phase. $\mathbb{Z}^+$ is a set of non-negative integers.

Action set for production control: $\mu(n, s) \in \mathbb{A}_p = [0, \mu_{max}]$ The production rate can be varied between 0 and μ_{max} and it is assumed that the maximum production rate is greater than the long run average arrival rate, i.e.,

$$\mu_{max} > \sum_{s \in \mathbb{S}} \hat{\pi}_s \sum_{i=1}^l \lambda_{i,s} \quad (1)$$

Note that we allow for $\lambda_{i,s}$ to be greater than μ_{max} for a certain phase s and class i , i.e. demand might exceed the available capacity for a particular season. Equation 1 is used to establish the existence of the optimal cost under the infinite horizon average expected cost criterion.

Action set for inventory rationing when class i demand arrives: $u_i(n, s) \in \mathbb{A}_r = \{0, 1\}$

- $u_i = 0$ Reject class i demand and incur penalty cost p_i
- $u_i = 1$ Satisfy class i demand

Transition probabilities: The transition probability $p(n', s' | n, s, \mu, u_i)$ for all $\mu \in [0, \mu_{max}]$, $u_i \in \{0, 1\}$ are expressed below.

$$\begin{aligned} p(n-1, s | n, s, \mu, u_i) &= \frac{u_i \lambda_{i,s}}{\nu} & \forall i \in \{1, \dots, l\}, \forall n \in \mathbb{N}, \forall s \in \mathbb{S} \\ p(n+1, s | n, s, \mu, u_i) &= \frac{\mu}{\nu} & \forall n \in \mathbb{Z}^+, \forall s \in \mathbb{S} \\ p(n, s' | n, s, \mu, u_i) &= \frac{Q_{ss'}}{\nu} & \forall n \in \mathbb{Z}^+, \forall s, s' \in \mathbb{S} \text{ and } s \neq s' \\ p(n, s | n, s, \mu, u_i) &= \left(1 - \frac{\sum_i u_i \lambda_{i,s} + \mu + \sum_{s' \in \mathbb{S}} Q_{ss'}}{\nu}\right) & \forall n \in \mathbb{Z}^+, \forall s, s' \in \mathbb{S} \text{ and } s \neq s' \end{aligned}$$

where $Q_{ss'}$ is the element of the generator matrix Q , $\nu = l\lambda_{max} + \eta + \mu_{max}$, $\lambda_{max} = \max_{i,s}\{\lambda_{i,s}\}$, and $\eta \geq \max\{-Q_{ss}|s \in \mathbb{S}\}$.

Cost function: The cost function consists of production cost, inventory holding cost and penalty cost. The inventory holding cost per unit time per unit of inventory is h . The production cost is assumed to be piecewise linear function of the production rate μ . More specifically, the production cost increases linearly with rate c_{reg} up to μ_{reg} where $0 < \mu_{reg} < \mu_{max}$. Subsequently, the production cost increases linearly with rate c_{max} . The production cost per unit time $c(\mu)$, is represented as

$$c(\mu) = \begin{cases} c_{reg}\mu & 0 \leq \mu \leq \mu_{reg} \\ c_{reg}\mu_{reg} + c_{max}(\mu - \mu_{reg}) & \mu_{reg} < \mu \leq \mu_{max} \end{cases} \quad (2)$$

where $0 < c_{reg} < c_{max}$ are scalar constants. Finally, we assume that $c_{reg} < p_l$, i.e. the penalty cost is high enough to justify production and satisfy demands of customers of any class. Note that the particular piece-wise linear cost structure described above is not a limiting feature of our model. We assume this structure for ease of illustration and our analysis can be extended to other general piece-wise linear and more general convex functions.

Within a transition, the total cost comprises of the immediate cost p_i , if arriving class i demand is rejected, and the cost that is incurred per unit time. For the k^{th} transition, the cost function per unit time is expressed as $C_k(n, s, \mu) = c(\mu) + h(n)$. The discounted cost per transition is then obtained by integrating the discount factor $e^{-\alpha t}$ and $C_k(n, s, \mu)$ over the inter-transition time and integrating the inter-transition time over its probability distribution. Thus, for the k^{th} transition, we have

$$\begin{aligned} C_{\alpha,k}(n, s, \mu) &= \sum_{i=1}^l p_i(1 - u_i) + \int_0^\infty \left(\int_0^\tau C_k(n, s, \mu) e^{-\alpha t} dt \right) \nu e^{-\nu \tau} d\tau \\ &= \sum_{i=1}^l p_i(1 - u_i) + \frac{1}{\alpha + \nu} C_k(n, s, \mu) \end{aligned}$$

Without loss of generality, we rescale the time to set $\alpha + \nu = 1$. We next formulate the objective function under finite horizon discounted cost criterion, infinite horizon discounted cost criterion, and infinite horizon average criterion.

3.1 Objective Function under Discounted Expected Cost Criterion

Let $v_{\alpha,k+1}(n, s)$ denote the finite horizon optimal discounted expected cost when $k + 1$ transitions are left. Then:

$$v_{\alpha,k+1}(n, s) = \inf_{\pi \in \Pi} \mathbb{E}_{(n,s)}^{\pi} \sum_{i=k+1}^1 C_{\alpha,i}(n', s', \mu_{\pi}) \quad \text{for all } n \in \mathbb{Z}^+ \text{ and for all } s \in \mathbb{S} \quad (3)$$

The corresponding optimality equation is written as:

$$\begin{aligned} v_{\alpha,k+1}(n, s) &= h(n) + \min_{\mu \in [0, \mu_{max}]} \left\{ c(\mu) + \mu v_{\alpha,k}(n+1, s) + (\mu_{max} - \mu) v_{\alpha,k}(n, s) \right\} \\ &+ \sum_{i=1}^l \lambda_{i,s} \min_{u_i \in \{0,1\}} \left\{ (1 - u_i(n, s))(v_{\alpha,k}(n, s) + p_i) + u_i(n, s) H_i v_{\alpha,k}(n, s) \right\} \\ &+ \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s}) v_{\alpha,k}(n, s) + \sum_{s' \in S, s \neq s'} Q_{ss'} v_{\alpha,k}(n, s') + (\eta + Q_{ss}) v_{\alpha,k}(n, s) \\ &\quad \text{for all } n \in \mathbb{Z}^+ \text{ and for all } s \in \mathbb{S} \end{aligned} \quad (4)$$

$$H_i v_{\alpha,k}(n, s) = \begin{cases} v_{\alpha,k}(n-1, s) & \text{if } n > 0 \\ v_{\alpha,k}(n, s) + p_i & \text{if } n = 0 \end{cases}$$

where $v_{\alpha,0}(n, s)$ is assumed 0 for all states. The first term in Equation 4 is the holding cost incurred during the $(k+1)^{st}$ transition. The next three terms are the cost-to-go functions corresponding to production, arrival, and phase transition respectively. The H operator ensures that the system incurs a penalty cost if it cannot satisfy an arriving demand due to the empty inventory buffer.

Next, consider the extension to the infinite horizon case. Denote $v_{\alpha}(n, s)$ as the infinite horizon optimal discounted expected cost function. Then,

$$\begin{aligned} v_{\alpha}(n, s) &= \inf_{\pi \in \Pi} \lim_{k \rightarrow \infty} \mathbb{E}_{(n,s)}^{\pi} \sum_{i=k+1}^1 C_{\alpha,i}(n', s', \mu_{\pi}) \\ &= \inf_{\pi \in \Pi} \lim_{k \rightarrow \infty} v_{\alpha,k+1}^{\pi}(n, s) \quad \text{for all } n \in \mathbb{Z}^+ \text{ and for all } s \in \mathbb{S} \end{aligned} \quad (5)$$

Denote $\lim_{k \rightarrow \infty} v_{\alpha, k+1}(n, s)$ as $v'_\alpha(n, s)$. Taking the limit $k \rightarrow \infty$ in Equation 4, we obtain

$$\begin{aligned}
v'_\alpha(n, s) &= h(n) + \min_{\mu \in [0, \mu_{max}]} \left\{ c(\mu) + \mu v'_\alpha(n+1, s) + (\mu_{max} - \mu) v'_\alpha(n, s) \right\} \\
&+ \sum_{i=1}^l \lambda_{i,s} \min_{u_i \in \{0,1\}} \left\{ (1 - u_i(n, s))(v'_\alpha(n, s) + p_i) + u_i(n, s) H_i v'_\alpha(n, s) \right\} \\
&+ \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s}) v'_\alpha(n, s) + \sum_{s' \in S, s \neq s'} Q_{ss'} v'_\alpha(n, s') + (\eta + Q_{ss}) v'_\alpha(n, s) \quad (6)
\end{aligned}$$

Since the state space is countably infinite and the cost function per transition is unbounded, $\lim_{k \rightarrow \infty} \inf_{\pi \in \Pi} v_{\alpha, k}^\pi(n, s)$ i.e., $v'_\alpha(n, s)$ need not exist, or, if it exists, it need not be equal to $\inf_{\pi \in \Pi} \lim_{k \rightarrow \infty} v_{\alpha, k+1}^\pi(n, s)$ i.e., $v_\alpha(n, s)$. Lemma 1 states that $v'_\alpha(n, s) = v_\alpha(n, s)$.

Lemma 1. $\lim_{k \rightarrow \infty} v_{\alpha, k}(n, s) = v_\alpha(n, s)$ for $n \in \mathbb{Z}^+$ and for $s \in \mathbb{S}$.

Lemma 1 follows from Theorem 11.5.3 of Puterman (1994). We restate the theorem in the context of our system and validate its assumptions for our system in A. Theorem 11.5.3 when applied to our system also guarantees the existence of a stationary deterministic optimal policy, and implies that any policy that minimizes the right side of Equation 6 is an optimal policy. In Section 4, we obtain the analytical expression of the optimal policy using this approach.

3.2 Objective Function under Average Expected Cost Criterion

For the average expected cost criterion, we only consider the infinite horizon case. Let g be the infinite horizon optimal average expected cost and $w(n, s)$ be the corresponding relative value function. Then:

$$g = \lim_{\alpha \downarrow 0} \alpha v_\alpha(n, s) \quad (7)$$

$$w(n, s) = \lim_{k \rightarrow \infty} (v_{\alpha_k}(n, s) - v_{\alpha_k}(n_0, s_0)) \quad (8)$$

where (n_0, s_0) is some distinguished state. The above expressions hold provided their limits exist. To obtain g and $w(n, s)$ we first subtract the discounted optimal cost $v_\alpha(n_0, s_0)$ from $v_\alpha(n, s)$ and subsequently take the limit $\alpha \rightarrow 0$. We get:

$$\begin{aligned}
& v_\alpha(n, s) - v_\alpha(n_0, s_0) \\
&= h(n) + \min_{\mu \in [0, \mu_{max}]} \left\{ c(\mu) - \alpha v_\alpha(n_0, s_0) + \mu[v_\alpha(n+1, s) - v_\alpha(n_0, s_0)] \right. \\
&\quad \left. + (\mu_{max} - \mu)[v_\alpha(n, s) - v_\alpha(n_0, s_0)] \right\} \\
&+ \sum_{i=1}^l \lambda_{i,s} \min_{u_i \in \{0,1\}} \left\{ (1 - u_i(n, s))(v_\alpha(n, s) - v_\alpha(n_0, s_0) + p_i) \right. \\
&\quad \left. + u_i(n, s)[H_i v_\alpha(n, s) - v_\alpha(n_0, s_0)] \right\} + \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s})[v_\alpha(n, s) - v_\alpha(n_0, s_0)] \\
&+ \sum_{s' \in S, s \neq s'} Q_{ss'}[v_\alpha(n, s') - v_\alpha(n_0, s_0)] + (\eta + Q_{ss})[v_\alpha(n, s) - v_\alpha(n_0, s_0)] \\
&\quad \text{for } n \in \mathbb{Z}^+ \text{ and } s \in \mathbb{S} \tag{9}
\end{aligned}$$

For systems with countable state space and unbounded reward, the limits in Equations 7 and 8 do not always hold. However, Theorem 1 of Sennott (1989) guarantees that the limits exist, there by providing the following relationship:

$$\begin{aligned}
w(n, s) + g &\geq h(n) + \min_{\mu \in [0, \mu_{max}]} \left\{ c(\mu) + \mu w_\alpha(n+1, s) + (\mu_{max} - \mu)w_\alpha(n, s) \right\} \\
&+ \sum_{i=1}^l \lambda_{i,s} \min_{u_i \in \{0,1\}} \left\{ (1 - u_i(n, s))(w_\alpha(n, s) + p_i) + u_i(n, s)H_i w_\alpha(n, s) \right\} \\
&+ \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s})w_\alpha(n, s) + \sum_{s' \in S, s \neq s'} Q_{ss'} w_\alpha(n, s') + (\eta + Q_{ss})w_\alpha(n, s) \\
&\quad \text{for } n \in \mathbb{Z}^+ \text{ and for } s \in \mathbb{S} \tag{10}
\end{aligned}$$

In B, we re-state Theorem 1 of Sennott (1989) and validate its assumptions in the context of our system. The theorem guarantees the existence of a stationary average expected cost optimal policy and also enables us to validate whether properties that hold under the discounted cost optimal policy also hold for the average cost optimal policy. The next section proves key properties of the optimal solution and optimal policy.

4 Properties of the Optimal Solution and Optimal Policy

We derive first and second order properties of the optimal cost function in Section 4.1, and using them we determine structural properties of the optimal policy in Section 4.2.

4.1 Properties of the Optimal Cost Function

Proposition 1. *For each $s \in \mathbb{S}$, $v_\alpha(n, s)$ is a convex function of n .*

An induction based proof of Proposition 1 is provided in C. The proposition implies that the first difference of the cost function $y_\alpha(n, s) = v_\alpha(n, s) - v_\alpha(n-1, s)$ is non-decreasing in n . Using the monotonicity of $y_\alpha(n, s)$ in n , we prove the following asymptotic result.

Proposition 2. *For a fixed $s \in \mathbb{S}$,*

1. $y_\alpha(n, s) \rightarrow \frac{h}{\alpha}$ as $n \rightarrow +\infty$
2. $y_\alpha(n, s) > -p_1$ for all $n \in \mathbb{Z}^+$

Part 1 of the Proposition 2 holds if we show $\frac{h}{\alpha}$ is the least upper bound of $y_\alpha(n, s)$, since $y_\alpha(n, s)$ is a non-decreasing function of n . In D, we first prove $\frac{h}{\alpha}$ is the upper bound of $y_\alpha(n, s)$ and then using contradiction argument, we show that $\frac{h}{\alpha}$ is the least upper bound of $y_\alpha(n, s)$. Part 2 of the Proposition 2 is proved directly using induction in D. Proposition 2 leads to Proposition 3 that yields another important property of $v_\alpha(n, s)$.

Proposition 3. *For each $s \in \mathbb{S}$, there exist a finite $n_{o,s} \in \mathbb{Z}^+$ such that $v_\alpha(n, s)$ is decreasing in n at the fixed s , for all $0 \leq n < n_{o,s}$ and increasing in n for all $n_{o,s} \leq n < \infty$.*

Proof. We prove by contradiction that $n_{o,s} < \infty$. If we assume that $n_{o,s}$ is not finite then for all $n \in \mathbb{Z}^+$, $y_\alpha(n, s) < 0$ since $y_\alpha(n, s)$ is non-decreasing function of n for each s . This implies $\frac{h}{\alpha}$ is not the least upper bound of $y_\alpha(n, s)$ since $0 < \frac{h}{\alpha}$. From Part 1 of Proposition 2, we infer that this is a contradiction. Hence, $n_{o,s} < \infty$. $\square$

4.2 Analytical Expression for the Optimal Policy

Recall that the policy which minimizes the right side of Equation 6 is an optimal policy. By minimizing the right side of Equation 6, we obtain a structured optimal policy which we state as Theorem 1.

Theorem 1. *The infinite horizon discounted cost optimal problem (Equation 6) has a structured optimal policy characterized by a season dependent inventory threshold vector $\mathbf{z}_s = [z_{1,s}, \dots, z_{l,s}, z_{\mu,s}, z_{b,s}]$ such that for each season s , it is optimal to:*

1. *Satisfy an arriving class i demand only if the inventory level is above $z_{i,s}$ for $i \in \{1, \dots, l\}$*
2. *Produce at rate μ_{max} if the inventory level drops below $z_{\mu,s}$*
3. *Produce at rate μ_{reg} if the inventory level drops below $z_{b,s}$ but is greater than or equal to $z_{\mu,s}$*
4. *Stop production if the inventory level equals or exceeds $z_{b,s}$*

The proof is provided in E. The optimal inventory allocation policy presented in Theorem 1 is of control limit type. The policy consists of phase dependent rationing level $z_{i,s}$ for each customer class i in season s . The optimal production control policy is also a phase (s) dependent base-stock policy with state (n, s) dependent production rate. Note that, the special structure of the production cost function expressed in Equation 2 leads to only two optimal production rates μ_{reg} and μ_{max} .

If we represent the season dependent inventory thresholds $\mathbf{z}_s$ as a column vector then the infinite horizon discounted expected cost optimal policy of the system for all seasons can be represented by the matrix Z_α as shown below:

$$Z_\alpha = \begin{bmatrix} z_{1,1} & \dots & z_{1,m} \\ \vdots & \ddots & \vdots \\ z_{l,1} & \dots & z_{l,m} \\ z_{\mu,1} & \dots & z_{\mu,m} \\ z_{b,1} & \dots & z_{b,m} \end{bmatrix}$$

The average cost optimal policy, represented by the optimal threshold matrix Z' , is obtained by taking the limit $Z_{\alpha_k} \rightarrow Z'$ where α_k is a subsequence of discount factors such that $\alpha_k \rightarrow 0$ (Sennott (1989)).

5 Optimal Policy for Systems with Service Level Constraints

In the next section, we analyze multi-class systems with constraints on service levels for the individual classes. Note that the optimal policies are obtained under the criterion of minimizing the long run average cost of the manufacturing system. However, optimal policies obtained under these settings might yield service levels (probability of meeting customer demands immediately from inventory) that are below the service level expectations for individual customer classes. In particular, individual classes of customer might impose constraints on service level, i.e. require that the system manage production and inventory so that the probability of lost sales for a particular class of customer is less than a desired level in each season. In this section, we consider the corresponding Markov decision problem (MDP) under the infinite horizon average cost criterion described in Section 3.2. We convert the MDP into its equivalent linear program and then add the service level constraint to the linear program. The problem $\mathbb{P}$ given below represents the corresponding linear program for the MDP problem analyzed in the previous section.

$$\mathbb{P} : \quad \text{Maximize} \quad g'$$

subject to:

$$g' + w(n, s) - \sum_{(n', s') \in \mathbb{Z}^+ \times \mathbb{S}} P(n', s' | n, s, \mu, u_i) w(n', s') \leq r(n, s, \mu, u_i) \quad (11)$$

$$\forall n \in \mathbb{Z}^+, s \in \mathbb{S}, \mu \in \mathbb{A}_p, u_i \in \mathbb{A}_r$$

$g', w(n, s)$ are unconstrained variables

where $r(n, s, \mu, u_i) = c(\mu) + h(n) + \sum_{i=1}^l \lambda_{i,s} (1 - u_i) p_i$.

Note that in the MDP we minimize the average cost, but in the primal linear program we maximize the variable g' . Due to the constraint 11 in the above primal linear program, the optimal average cost g is an upper bound of the objective variable g' of the primal linear program. Hence, we maximize the objective function g' to attain the optimal average cost.

Also, the action space $\mathbb{A}_p$ is continuous and needs to be discretized for linear programming formulation. Since, the optimal production rate are either 0, μ_{reg} , or μ_{max} , we consider a set of possible discrete production rates μ_i for $i = 1, \dots, n$ such that $0 \leq \mu_i \leq \mu_{max}$ so that $\mathbb{A}_p = \{0, \dots, \mu_{reg}, \dots, \mu_{max}\}$. We next formulate the dual linear program since the

dual linear program has less constraints than the primal linear program. Another advantage of formulating the dual linear program is that we can directly estimate the probability of the system being in state (n, s) and action μ being chosen. We also add the service level constraint to the following dual linear program formulation.

Dual Linear Program

$$\text{Minimize} \quad \sum_{u_i \in \mathbb{A}_r} \sum_{\mu \in \mathbb{A}_p} \sum_{(n,s) \in \mathbb{Z}^+ \times \mathbb{S}} r(n, s, \mu, u_i) x(n, s, \mu, u_i)$$

subject to:

$$\sum_{u_i \in \mathbb{A}_r} \sum_{\mu \in \mathbb{A}_p} x(n', s', \mu, u_i) - \sum_{u_i \in \mathbb{A}_r} \sum_{\mu \in \mathbb{A}_p} \sum_{(n,s) \in \mathbb{Z}^+ \times \mathbb{S}} P(n', s' | n, s, \mu, u_i) x(n, s, \mu, u_i) = 0 \quad \forall n' \in \mathbb{Z}^+, s' \in \mathbb{S} \quad (12)$$

$$\sum_{u_i \in \mathbb{A}_r} \sum_{\mu \in \mathbb{A}_p} \sum_{(n,s) \in \mathbb{Z}^+ \times \mathbb{S}} x(n, s, \mu, u_i) = 1 \quad (13)$$

$$\sum_{u_j \in \mathbb{A}_r} \sum_{\mu \in \mathbb{A}_p} \sum_{n \in \mathbb{Z}^+} x(n, s, \mu, u_i = 1, u_j) / \hat{\pi}_s \geq SL_i \lambda_{i,s} \quad \forall i \neq j \in \{1, \dots, l\}, s \in \mathbb{S} \quad (14)$$

$$x(n, s, \mu, u_i) \geq 0 \quad \forall n \in \mathbb{Z}^+, s \in \mathbb{S}, \mu \in \mathbb{A}_p, u_i \in \mathbb{A}_r$$

where $x(n, s, \mu, u_i)$ is the probability that the system is in state (n, s) and chooses action μ, u_i and SL_i is the service level of class i customer. In the dual linear program, the constraint 12 is interpreted as the flow balance equation for each state, the constraint 13 ensures the probabilities sum up to one, and the constraint 14 is the service level constraint conditioned on each season for each customer class.

Usually a constrained MDP problem has a randomized policy. However, in this case, the optimal action is randomized only at the state with inventory level such that the optimal production rate switches from μ_{max} to μ_{reg} . Since the optimal action is randomized only for one of the several states, we ignore the randomness in the action and set the optimal production rate as μ_{reg} at that particular inventory level. Thus, under this setting, the optimal policy for the system under service level constraint retain the same structure as the one without service level constraints.

The optimal policy, specified by the threshold matrix Z_α , has a simple structure that could be used in practice. The threshold matrix could be determined by solving the above formulated linear program. The production and cost parameters required for the linear program could

be adjusted according to a company's operating conditions, whereas the demand parameters could be determined by fitting the observed demands to the MMPP using regression-based techniques. The service level requirements of the customers could be incorporated by adding them as constraints to the linear program.

The optimal policy is best used as a guiding principle for production-inventory control decisions. For instance, it could be used to obtain ball-park estimates of production rate and amount of inventory to ration among different customers for a given demand scenario, and also to benchmark existing production-inventory control policies.

6 Numerical Studies

We conduct numerical studies to study the impact of different seasonal patterns (Section 6.1) as well as the impact of service level constraints on optimal policy and cost (Section 6.2).

6.1 Impact of Different Seasonal Demand Patterns

In this section, we conduct a numerical study to demonstrate the monotone properties of the optimal policy and analyze the impact of different patterns of customer demands on the optimal policy and cost. We consider a manufacturing system with two classes of customers and demands having three seasons. The generator matrix Q for the corresponding MMPP is given below

$$Q = \begin{bmatrix} -q & q & 0 \\ q & -2q & q \\ 0 & q & -q \end{bmatrix}$$

where q is set equal to 0.01 implying that the average duration of each season is 100 days. Note that, the structure of matrix Q is such that, the phase transition process is a stochastically monotone process. We consider two classes of customers such that the penalty cost of class 1 customer is 10 times that of class 2 customers. The regular production rate μ_{reg} is 10 units per day, and the maximum production rate μ_{max} is 15 units per day respectively. The holding cost h , production cost rate c_{reg}, c_{max} , penalty cost p_i for classes $i \in \{1, 2\}$ are listed in Table 1.

We consider two cases, corresponding to synchronized demands (SD) and complementary demands (CD), respectively. In the case of synchronized demands, the average demand for each customer class is 4, 6, and 8 units per day in seasons 1, 2, and 3 respectively. This implies

Table 1: Cost parameters

Holding cost	$h = 1$
Production cost	$c_{reg} = 10, c_{max} = 20$
Class 2 penalty cost	$p_2 = 1.1 \times c_{max}$
Class 1 penalty cost	$p_1 = 10 \times p_2$

that the total average demand in each season is 8, 12, and 16 units per day respectively. In the case of complementary demands, the average demand for class 1 customers is 4, 6, and 8 units per day in seasons 1, 2, and 3 respectively, while the average demand for class 2 customers is 8, 6, and 4 units per day in seasons 1, 2, and 3 respectively. This implies that the total average demand is leveled across the seasons and is equal to 12 units per day in each of the three seasons. The average cost and the threshold levels of the policies are estimated using policy iteration algorithm (Puterman (1994)). For computational purposes, we truncate the state space to $[0:1200]$. The optimal threshold levels and the optimal cost for both the SD and CD cases are listed in Table 2.

Table 2: Comparing systems with different seasonal demand patterns

Cases	Demand $\begin{bmatrix} \lambda_{11} & \lambda_{12} & \lambda_{13} \\ \lambda_{21} & \lambda_{22} & \lambda_{23} \end{bmatrix}$	Optimal Cost g	Optimal Policy $\begin{bmatrix} z_{1,1} & z_{1,2} & z_{1,3} \\ z_{2,1} & z_{2,2} & z_{2,3} \\ z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$
Synchronized demands	$\begin{bmatrix} 4 & 6 & 8 \\ 4 & 6 & 8 \end{bmatrix}$	4.863	$\begin{bmatrix} 0 & 0 & 0 \\ 2 & 5 & 8 \\ 3 & 7 & 14 \\ 12 & 33 & 62 \end{bmatrix}$
Complementary demands	$\begin{bmatrix} 4 & 6 & 8 \\ 8 & 6 & 4 \end{bmatrix}$	4.640	$\begin{bmatrix} 0 & 0 & 0 \\ 3 & 5 & 7 \\ 6 & 8 & 9 \\ 31 & 33 & 34 \end{bmatrix}$

In Table 2, we first discuss the results for the case with synchronized demands. We observe that the class 1 rationing level ($z_{1,s}$) is zero in all seasons implying that class 1 customers are satisfied whenever the inventory buffer has produced good inventory. Further, the class 2 rationing level ($z_{2,s}$) increases as we move from low demand season (season 1) to the peak demand season (season 3), implying that more inventory is reserved to satisfy class 1 customers in seasons 2 and 3. The last two rows list the production threshold levels ($z_{\mu,s}$ and $z_{b,s}$). For instance, in season 3, the optimal policy suggest to switch to maximum production rate μ_{max} whenever the inventory level drop below 14, and to switch to regular production

rate, μ_{reg} between inventory levels 14 and 62, and stop the production when the inventory is 62 units. Note that both $z_{\mu,s}$ and $z_{b,s}$ increase monotonically with the season, and the difference between these thresholds demonstrates the value of capacity flexibility. For instance, note that $z_{\mu,s}$ increases from 3 to 14 as the demand increases from 4 to 8. To respond to the increased demand in season 3, the system leverages the capacity flexibility to operate the system at a higher capacity and meet demands.

Next, we compare the thresholds for cases with synchronized (SD) and complementary (CD) demands. We observe that for the case of complementary demands too, the class 2 rationing level ($z_{2,s}$) increases from season 1 to season 3, even though the total average demand is the same across seasons. This is because, the proportion of class 1 customers is higher in season 3 than in season 1. Next, we compare the production threshold levels ($z_{\mu,s}$ and $z_{b,s}$) for the two cases. We observe that for the SD case, $z_{b,s}$ increases from 12 to 62 to account for the increased demand in the peak season, while in the CD case, $z_{b,s}$ varies from 31 to 34 since the total average demand is same across seasons. Further, the optimal average cost for the SD case is 5% higher compared to the cost of the CD case. This increase in cost is because when the total average demand in a season exceeds the regular production rate, the optimal policy recommends that the system switch to the maximum production rate at relatively higher values of the inventory.

6.2 Impact of Service Level on Optimal Cost and Policy

We next investigate the service level that the customer receives when the manufacturer runs the system using the optimal policy. Let $SL_{i,s}$ denote the service level for class i customers in season s where service level is measured as the percentage of demands satisfied by the manufacturer from stock. Table 3 lists the service level for the two classes of customers under the synchronized demand and complementary demand cases.

Table 3: Service levels for synchronized and complementary demands

Case	Season 1	Season 2	Season 3
Synchronized Demand, Class 1	99.89	99.92	99.92
Synchronized Demand, Class 2	97.90	86.40	73.48
Complementary Demand, Class 1	99.88	99.94	99.90
Complementary Demand, Class 2	91.48	89.63	83.10

From Table 3 we observe that the service level of class 2 customers are lower than that of class 1 customers, and the difference in service levels is highest in the season with peak demand. This suggests that although the optimal policy identified in Section 4 minimizes

total cost, it may result in service levels that are not acceptable for class 2 customers. Next, we use the formulation in Section 5 to analyze the impact of service level constraints on the optimal cost and policy. Table 4 lists the optimal cost and the optimal policy under two scenarios, (i) when the service level constraints are 99.99% for class 1 customers and 95% for class 2 customers, and (ii) when the service level constraints are 99.99% for class 1 customers and 98% for class 2 customers. As expected, the optimal cost increases when these constraints are imposed on the system. The optimal cost further increases when the service level constraint for class 2 customers is increased from 95% to 98%.

Table 4: Impact of service level on optimal policy and cost

		Class 1: 99.99%, Class 2: 95%		Class 1: 99.99%, Class 2: 98%	
Cases	Demand	Optimal Cost	Optimal Policy	Optimal Cost	Optimal Policy
	$\begin{bmatrix} \lambda_{11} & \lambda_{12} & \lambda_{13} \\ \lambda_{21} & \lambda_{22} & \lambda_{23} \end{bmatrix}$	g	$\begin{bmatrix} z_{1,1} & z_{1,2} & z_{1,3} \\ z_{2,1} & z_{2,2} & z_{2,3} \\ z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$	g	$\begin{bmatrix} z_{1,1} & z_{1,2} & z_{1,3} \\ z_{2,1} & z_{2,2} & z_{2,3} \\ z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$
Synchronized Demands	$\begin{bmatrix} 4 & 6 & 8 \\ 4 & 6 & 8 \end{bmatrix}$	6.964	$\begin{bmatrix} 0 & 0 & 0 \\ 3 & 47 & 8 \\ 4 & 170 & 269 \\ 18 & 223 & 288 \end{bmatrix}$	10.290	$\begin{bmatrix} 0 & 9 & 0 \\ 76 & 105 & 7 \\ 96 & 297 & 393 \\ 155 & 352 & 419 \end{bmatrix}$
Complementary Demands	$\begin{bmatrix} 4 & 6 & 8 \\ 8 & 6 & 4 \end{bmatrix}$	4.710	$\begin{bmatrix} 0 & 0 & 0 \\ 4 & 6 & 8 \\ 10 & 13 & 16 \\ 35 & 37 & 40 \end{bmatrix}$	4.765	$\begin{bmatrix} 0 & 0 & 0 \\ 3 & 5 & 7 \\ 13 & 15 & 19 \\ 38 & 40 & 43 \end{bmatrix}$

In Table 4, we observe that the cost for the CD case is lower than that for the SD case. When the demands of the two customer classes are synchronized (SD), achieving the desired the service level for both classes of customers is challenging, especially in the peak demand season when the total average demand exceeds the maximum production rate. However, when the demand patterns are complementary (CD), achieving the service level constraint for both classes is less challenging, relatively, since the total average demand in a season remains the same across seasons and is always less than the maximum production rate. The increase in cost to achieve higher service levels is marginal in the CD case as compared to the SD case. Understanding such interactions between seasonal demand patterns and service level constraints can be very beneficial to the practitioner.

Next, we compare the optimal threshold levels for these cases to those computed in Table 2. When the service level constraints are imposed, for the SD case, the class 2 rationing levels ($z_{2,s}$) increase. However, the increase in these rationing levels demonstrates a phenomenon where the system tends to build-in-advance. Comparing values of $z_{2,s}$ for the SD case in Table 2 and 4, we note that $z_{2,s} = [2 \ 5 \ 8]$ in Table 2, $[3 \ 47 \ 8]$ (in Table 4) when

$SL_{2,s} = 95\%$ and $[74 \ 105 \ 7]$ (in Table 4) when $SL_{2,s} = 98\%$. The increase in $z_{2,1}$ and $z_{2,2}$ is because under the optimal policy, the system reserves additional inventory in seasons 1 and 2 to be able to meet service level requirements for class 2 customers in season 3. In fact, we observe that the class 1 rationing level in season 2 is greater than zero, suggesting that under the optimal policy, in season 2, the system chooses to reject class 1 customer demands when the inventory level is below 10 so that it can reserve inventory to satisfy demands for for class 1 and 2 customers in season 3. The corresponding increase in the production thresholds $z_{\mu,s}$ and $z_{b,s}$ further emphasize the build-in-advance effect driven by the service level constraints. Understandably, this build in advance phenomenon is more pronounced in the SD demand case. For the CD case, the increased service level requirements do drive increases in thresholds, but only marginally. This contrast emphasizes the joint impact of seasonal trends and service level constraints on optimal policy.

7 Concluding Remarks

In this paper, we analyzed a manufacturing system that has the ability to leverage capacity flexibility to satisfy seasonal demands of its customers. The manufacturing system caters to multiple customer classes that differ in terms of their penalty for lost sales. The system dynamics is analyzed as a Markov decision process where the objective is to minimize the total production-inventory cost under both the infinite horizon discounted expected cost and average expected cost criteria. We establish the existence of the optimal solution and derive an analytical expression for the optimal policy. We believe the inventory threshold-level based structure of the optimal policy is conducive for implementation in real world practice.

We then analyze the impact of service level constraints on the optimal policy and the optimal cost under different demand patterns. When the demand patterns are synchronized, the optimal cost and the optimal threshold levels increase significantly under the service level constraints compared to the case when the seasonal demand patterns are complementary. Further, for satisfying the service level constraints of the customers in the peak demand season, the optimal policy not only suggest to build up inventories for the peak demand season, but also to reserve inventories and not satisfy customers of any classes at low inventory levels in the season preceding the peak demand season, especially when the seasonal demand patterns of the customers are synchronized. These insights on the impact of the service level constraints on the optimal policy and cost can be very beneficial to managers as they decide optimal policies under different seasonal demand settings.

We would like to point out two key directions for future research. First, the assumption of complete visibility of season change could be relaxed. In practice, the change in the demand seasons are not completely observable to manufacturers. Under such condition, the manufacturers might wait further to gain more information on the demand season before adjusting their production and inventory levels. It would be interesting to characterize the structure of the optimal policy under the partially-observed demand seasons. Second, the research could be extended from single product to multiple-product manufacturing systems to develop resource allocation policies among different products with distinct seasonal demand patterns.

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A Validating Assumptions for Theorem 11.5.3 of Puterman (1994) for our System

1. The optimality equation has a unique solution in $\mathbb{V}_\omega$ which equals v_α .
2. If a policy minimizes the right side of Equation 6, then it is a stationary deterministic optimal policy.
3. There exist an optimal stationary deterministic policy whenever the action set is either finite or compact set, and $C_{\alpha,k}(n, s, \mu, u_i)$ and $P(n', s'|n, s, \mu, u_i)$ is a continuous function of μ and for all $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$.
4. For each $v_{\alpha,0} \in \mathbb{V}_w$, $v_{\alpha,k} = \mathcal{L}^k v_{\alpha,0}$ converges to v_α

if assumptions 1, 2a, 2b, and 3 hold for our system.

Assumption 1: There exist a constant $c' < \infty$ such that

$$\sup_{\mu \in [0, \mu_{max}], u_i(n,s) \in \{0,1\}} |hn + c(\mu) + \sum_{i=1}^l \frac{\lambda_i}{\nu} p_i u_i(n, s)| \leq c' \omega(n, s)$$

$\omega(n, s)$ is any arbitrary positive real valued function of $\mathbb{Z}^+ \times \mathbb{S}$ satisfying $\inf_{n \in \mathbb{Z}^+, s \in \mathbb{S}} \omega(n, s) > 0$

Choose $\omega(n, s) = hn + \mu_{max} c_{max} + p_1$. Assumption 1 then holds for $c' = 1$.

Assumption 2a: There exist a constant $\kappa : 0 \leq \kappa < \infty$ for which

$$\sum_{n \in \mathbb{Z}^+, s \in \mathbb{S}} P(n', s' | n, s, \mu, u_i) \omega(n', s') \leq \kappa \omega(n, s) \quad (15)$$

for $\mu \in [0, \mu_{max}]$, $u_i \in \{0, 1\}$, $i \in \{0, \dots, l\}$ and for $n \in \mathbb{Z}^+$ and $s \in \mathbb{S}$

Assumption 2b: For each r , $0 \leq r < 1$, there exists an β , $0 \leq \beta \leq 1$ and an integer J such that

$$r^J \sum_{n' \in \mathbb{Z}^+, s \in \mathbb{S}} P_\pi^J(n', s' | n, s) \omega(n, s) \leq \beta \omega(n, s)$$

for all $\pi = (d_1, \dots, d_J)$ where $d_k \in D^{MD}$; $1 \leq k \leq J$

Consider the left hand side of Equation 15.

$$\begin{aligned}
\sum_{n \in \mathbb{Z}^+, s \in \mathbb{S}} P(n', s' | n, s, \mu, u_i) \omega(n', s') &= \mu_{\max} c_{\max} + h \left[n \left(\frac{\nu - \sum_{i=1}^l \lambda_{i,s} u_i - \mu + \sum_{s' \in \mathbb{S}} Q_{s,s'}}{\nu} \right) \right. \\
&\quad \left. + (n-1) \sum_{i=1}^l \frac{\lambda_{i,s}}{\nu} u_i + (n+1) \frac{\mu}{\nu} \right] + p_1 \\
&\leq \mu_{\max} c_{\max} + hn + h + p_1 \\
&= \omega(n, s) + h
\end{aligned} \tag{16}$$

Proposition 6.10.5 of Puterman (1994) states that suppose there exist a constant $L > 0$ for which

$$\sum_{n \in \mathbb{Z}^+, s \in \mathbb{S}} P(n', s' | n, s, a) \omega(n', s') \leq \omega(n, s) + L$$

for $a \in A_{n,s}$ and $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$, then Assumptions 2a and 2b hold. Assumptions 2a and 2b hold by setting $L = h$ and from Proposition 6.10.5 of Puterman (1994).

Assumption 3: $\nu < \infty$

Since $\lambda_{\max}, \mu_{\max}, \eta$ are finite, Assumption 4 holds.

Note that the conditions mentioned in part 3 of Theorem 11.5.3 can easily be validated for our system. The action set $[0, \mu_{\max}]$ is a compact set, since $\mu_{\max} < \infty$, and the inventory allocation decision set is finite. Also, the cost function $C_{\alpha,k}(n, s, \mu)$ and the probability function for the production term, $\frac{\mu}{\nu}$ are continuous functions of μ .

B Validating Assumptions for Theorem 1 of Sennott (1989) for our System

Theorem 1 of Sennott (1989) states that: If Equation 1 holds then

1. *There exist a constant $g^* = \lim_{\alpha \downarrow 0} \alpha v_{\alpha}(n, s)$ and a vector $w(n, s) = \lim_{k \rightarrow \infty} w_{\alpha_k}(n, s) := v_{\alpha_k}(n, s) - v_{\alpha_k}(n_0, s_0)$ for a subsequence $\alpha_k \downarrow 0$ such that $(g^*, w(n, s))$ satisfy the Inequality 10 for $n \in \mathbb{Z}^+$ and $s \in \mathbb{S}$, where (n_0, s_0) is some distinguished state.*
2. *Any stationary policy that minimizes the right side of Inequality 10 is the optimal policy.*

3. There exists a stationary average expected cost optimal policy $\mu^*(n, s)$ that is a limit point of a sequence of discounted expected cost optimal policies $\mu_{\alpha_k}^*(n, s)$, i.e., $\mu_{\alpha_k}^*(n, s) \rightarrow \mu^*(n, s)$ as $\alpha_k \rightarrow 0$

if assumptions 4, 5, 6, and 7 (stated below) hold for our system.

Assumption 4: For each $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$, $0 \leq C(n, s, \mu) < \infty$

Assumption 4 holds since the cost function is non-negative and is finite for each (n, s) .

Assumption 5: $v_\alpha(n, s) < +\infty$ for each $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$

Using Assumptions 1, 2a, and 2b of A, it can be shown that

$$v_\alpha^\pi(n, s) \leq (1 + r + \dots + (r\kappa)^{J-1}) \frac{\omega(n, s)}{1 - \beta} \text{ for all } \pi \in \Pi^{MR}$$

where Π^{MR} is a set of Markov randomized policies. Since $\omega(n, s) < \infty$ for each (n, s) , we have $v_\alpha^\pi(n, s) < \infty$. Thus, $v_\alpha(n, s) \leq v_\alpha^\pi(n, s) < \infty$.

Assumption 6: There exist a non-negative K such that $-K \leq w_\alpha(n, s)$ for $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$.

From Proposition 3, $v_\alpha(n, s)$ attains minimum at $(n_{0,s}, s)$ for each $s \in \mathbb{S}$.

Let $(n_0, s_0) = \arg \min\{(v_\alpha(n_{0,1}, 1), \dots, (v_\alpha(n_{0,m}, m))\}$, and $w_\alpha(n, s) = v_\alpha(n, s) - v_\alpha(n_0, s_0)$, then $w_\alpha(n, s) \geq 0$ for $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$. Then Assumption 6 holds by setting $K = 0$.

Assumption 7: There exist an α_0 and non-negative number $M(n, s)$ such that

1. $M(n, s) < \infty$
2. For each $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$, $w_\alpha(n, s) \leq M(n, s)$ for $0 \leq \alpha < \alpha_0$
3. For each $(n, s) \in \mathbb{Z}^+ \times \mathbb{S}$ there exist $\mu_{n,s} \in [0, \mu_{max}]$ and $u_{i,n,s}$ for all $i \in \{1, \dots, l\}$

$$\sum_{n', s' \in \mathbb{Z}^+ \times \mathbb{S}} P(n', s' | n, s, \mu_{n,s}, u_{i,n,s}) M(n', s') < \infty \quad (17)$$

We validate Assumption 7 using Proposition 5 of Sennott (1989). The proposition 5 of Sennott (1989) holds if there exist a stationary policy π that induces an irreducible, ergodic Markov chain and its long-run average expected cost $g^\pi < \infty$. We consider a policy π such that $\mu_{n,s} = \mu_{max}$ for $n < N_0$ and $\mu_{n,s} = 0$ for $n \geq N_0$ for all $s \in \mathbb{S}$, where N_0 is a positive integer; with a first come first serve demand fulfillment policy among different classes of customers. Then using arguments similar to Proposition 2.2 of Kumar et al. (2013), it can

be shown that under the condition described in Equation 1, the Markov chain induced due to this policy is irreducible, ergodic in the state space $[0, N_0] \times \mathbb{S}$, and has finite long-run expected average cost. Note that we have validated Assumption 7 under finite upper bound of the net inventory level. This condition is not restrictive for practical applications, since the inventory levels are always finite.

C Proof of Proposition 1

Proposition. For each $s \in \mathbb{S}$, $v_\alpha(n, s)$ is a convex function of n .

Proof. Using induction, we first show $y_{\alpha,k}(n, s)$ is non-decreasing in n and then extend the result to $y_\alpha(n, s)$ using Lemma 1. For $k = 0$, the result holds since $v_{\alpha,0}(n, s) = 0$ for all states. For the inductive step, assume $v_{\alpha,k}(n, s)$ is a convex function of n for some k , i.e. $y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s) \geq 0$ for $n \in \mathbb{Z}^+$, for each $s \in \mathbb{S}$. We now prove $y_{\alpha,k+1}(n, s) - y_{\alpha,k+1}(n-1, s) \geq 0$. Let $\mu^*(n, s)$ denote the optimal production rate at state (n, s) , and let $u_i^*(n, s)$ denote the optimal inventory allocation decision for a class i arrival in state (n, s) . In this proof, since we show $y_{\alpha,k}(n, s)$ is non-decreasing in n for a fixed s , we suppress s while denoting the optimal policy. Hence, we denote $\mu^*(n, s)$ as μ_n^* and $u_i^*(n, s)$ as $u_{i,n}^*$. We have

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) - y_{\alpha,k+1}(n-1, s) \\
&= h + \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n}^*)(p_i + v_{\alpha,k}(n, s)) + u_{i,n}^* H_i v_{\alpha,k}(n, s) \right\} \\
&\quad - \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-1}^*)(p_i + v_{\alpha,k}(n-1, s)) + u_{i,n-1}^* H_i v_{\alpha,k}(n-1, s) \right\} \\
&\quad + \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s}) (v_{\alpha,k}(n, s) - v_{\alpha,k}(n-1, s)) \\
&\quad + \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} (v_{\alpha,k}(n, s') - v_{\alpha,k}(n-1, s')) + (\eta + Q_{ss}) (v_{\alpha,k}(n, s) - v_{\alpha,k}(n-1, s)) \\
&\quad + c(\mu_n^*) + \mu_n^* v_{\alpha,k}(n+1, s) + (\mu_{max} - \mu_n^*) v_{\alpha,k}(n, s) \\
&\quad - \left[c(\mu_{n-1}^*) + \mu_{n-1}^* v_{\alpha,k}(n, s) + (\mu_{max} - \mu_{n-1}^*) v_{\alpha,k}(n-1, s) \right] \\
&\quad - \left[h + \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-1}^*)(p_i + v_{\alpha,k}(n-1, s)) + u_{i,n-1}^* H_i v_{\alpha,k}(n-1, s) \right\} \right. \\
&\quad \left. - \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-2}^*)(p_i + v_{\alpha,k}(n-2, s)) + u_{i,n-2}^* H_i v_{\alpha,k}(n-2, s) \right\} \right]
\end{aligned}$$

$$\begin{aligned}
& + \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s}) (v_{\alpha,k}(n-1, s) - v_{\alpha,k}(n-2, s)) \\
& + \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} (v_{\alpha,k}(n-1, s') - v_{\alpha,k}(n-2, s')) \\
& + (\eta + Q_{ss}) (v_{\alpha,k}(n-1, s) - v_{\alpha,k}(n-2, s)) \\
& + c(\mu_{n-1}^*) + \mu_{n-1}^* v_{\alpha,k}(n, s) + (\mu_{max} - \mu_{n-1}^*) v_{\alpha,k}(n-1, s) \\
& - \left[c(\mu_{n-2}^*) + \mu_{n-2}^* v_{\alpha,k}(n-1, s) + (\mu_{max} - \mu_{n-2}^*) v_{\alpha,k}(n-2, s) \right] \quad (18)
\end{aligned}$$

Combining terms in Equation 18 with the same coefficients yields:

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) - y_{\alpha,k+1}(n-1, s) \\
& = \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n})(p_i + v_{\alpha,k}(n, s)) + u_{i,n} H_i v_{\alpha,k}(n, s) \right\} \\
& - \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-1})(p_i + v_{\alpha,k}(n-1, s)) + u_{i,n-1} H_i v_{\alpha,k}(n-1, s) \right\} \\
& - \left[\sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-1})(p_i + v_{\alpha,k}(n-1, s)) + u_{i,n-1} H_i v_{\alpha,k}(n-1, s) \right\} \right. \\
& \left. - \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-2})(p_i + v_{\alpha,k}(n-2, s)) + u_{i,n-2} H_i v_{\alpha,k}(n-2, s) \right\} \right] \\
& + \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s} + \eta + Q_{ss}) (y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s)) \\
& + \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} (y_{\alpha,k}(n, s') - y_{\alpha,k}(n-1, s')) \\
& + c(\mu_n^*) + \mu_n^* y_{\alpha,k}(n+1, s) - c(\mu_{n-1}^*) + (\mu_{max} - \mu_{n-1}^*) y_{\alpha,k}(n, s) \\
& - c(\mu_{n-1}^*) - \mu_{n-1}^* y_{\alpha,k}(n, s) + c(\mu_{n-2}^*) - (\mu_{max} - \mu_{n-2}^*) y_{\alpha,k}(n-1, s) \quad (19)
\end{aligned}$$

Now, we show right side of Equation 19 is non-negative. The sum of terms in the first four lines of right side of Equation 19 is non-negative, and the non-negativity can be shown along the lines of proof of Lemma 1 of Ha (1997a). The term in the fifth and sixth lines are non-negative due to induction assumption. Now, consider the terms in the last two lines of right side of Equation 19. Replace the optimal action μ_{n-1}^* in the seventh line of the right side of Equation 19 by the suboptimal action μ_n^* ; and the optimal action μ_{n-1}^* in the eight line of right side of Equation 19 by the suboptimal action μ_{n-2}^* .

$$\mu_n^* (y_{\alpha,k}(n+1, s) - y_{\alpha,k}(n, s)) + (\mu_{max} - \mu_{n-2}^*) (y_{\alpha,k}(n, s) - y_{\alpha,k}(n-1, s)) \quad (20)$$

Each of the two terms in Equation 20 are positive due to the induction assumption. Thus $v_{\alpha,k+1}(n, s)$ is a convex function of n for each $s \in \mathbb{S}$. Finally, using Lemma 1, we conclude $v_{\alpha}(n, s)$ is a convex function of n , i.e. $y_{\alpha}(n, s)$ is non-decreasing in n for a given $s \in \mathbb{S}$. $\square$

D Proof of Proposition 2

Proposition. For a fixed $s \in \mathbb{S}$,

1. $y_{\alpha}(n, s) \rightarrow \frac{h}{\alpha}$ as $n \rightarrow +\infty$
2. $y_{\alpha}(n, s) > -p_1$ for all $n \in \mathbb{Z}^+$

Proof. First we show $\frac{h}{\alpha}$ is an upper bound and then using contradiction argument, we prove $\frac{h}{\alpha}$ is the least upper bound. To show $\frac{h}{\alpha}$ is an upper bound, we first show $\frac{h}{\alpha}$ is an upper bound for the finite horizon case using induction i.e. $y_{\alpha,k}(n, s) \leq \frac{h}{\alpha}$ for all $k \geq 0$ and then using Lemma 1, extend the result to infinite horizon. Note that $y_{\alpha,0}(n, s) \equiv 0$ for $n \in \mathbb{Z}^+$ and $s \in \mathbb{S}$. Assume for a given k , $y_{\alpha,k}(n, s) \leq \frac{h}{\alpha}$. We show below $y_{\alpha,k+1}(n, s) \leq \frac{h}{\alpha}$. We denote the optimal inventory allocation decision $u_i^*(n, s)$ as $u_{i,n}$ and the optimal production rate $\mu_i^*(n, s)$ as μ_n^*

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) \\
&= h + \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n})(p_i + v_{\alpha,k}(n, s)) + u_{i,n} H_i v_{\alpha,k}(n, s) \right\} \\
&\quad - \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-1})(p_i + v_{\alpha,k}(n-1, s)) + u_{i,n-1} H_i v_{\alpha,k}(n-1, s) \right\} \\
&\quad + \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s})(v_{\alpha,k}(n, s) - v_{\alpha,k}(n-1, s)) \\
&\quad + \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} (v_{\alpha,k}(n, s') - v_{\alpha,k}(n-1, s')) + (\eta + Q_{ss})(v_{\alpha,k}(n, s) - v_{\alpha,k}(n-1, s)) \\
&\quad + c(\mu_n^*) + \mu_n^* v_{\alpha,k}(n+1, s) + (\mu_{max} - \mu_n^*) v_{\alpha,k}(n, s) \\
&\quad - \left[c(\mu_{n-1}^*) + \mu_{n-1}^* v_{\alpha,k}(n, s) + (\mu_{max} - \mu_{n-1}^*) v_{\alpha,k}(n-1, s) \right] \tag{21}
\end{aligned}$$

Replacing the optimal μ_n^* by the suboptimal μ_{n-1}^* , replacing the optimal $u_{i,n}$ by the suboptimal $u_{i,n-1}$ in the Equation 21, and assuming $n > 0$, we obtain the following inequality:

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) \\
& \leq h + \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n-1})(y_{\alpha,k}(n, s)) + u_{i,n-1}y_{\alpha,k}(n-1, s) \right\} \\
& + \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s})y_{\alpha,k}(n, s) \sum_{s' \neq s \in \mathbb{S}} Q_{ss'} y_{\alpha,k}(n, s') + (\eta + Q_{ss})(y_{\alpha,k}(n, s)) \\
& + \mu_{n-1}^* y_{\alpha,k}(n+1, s) + (\mu_{max} - \mu_{n-1}^*)y_{\alpha,k}(n, s)
\end{aligned} \tag{22}$$

Replacing $y_{\alpha,k}(n, s)$ by its upper bound $\frac{h}{\alpha}$ in Equation 22, we obtain,

$$\begin{aligned}
y_{\alpha,k+1}(n, s) & \leq h + (l\lambda_{max} + \eta + \mu_{max})\frac{h}{\alpha} \\
& = (\alpha + \nu)\frac{h}{\alpha} \\
& = \frac{h}{\alpha} \quad \text{since } \alpha + \nu = 1
\end{aligned} \tag{23}$$

Thus, $\frac{h}{\alpha}$ is an upper bound of $y_{\alpha,k}(n, s)$ for $k \geq 0$. Next, we show by contradiction that $\frac{h}{\alpha}$ is the least upper bound of $y_{\alpha,k}(n, s)$. Let $(\frac{h'}{\alpha})$ be the least upper bound of $y_{\alpha,k}(n, s)$ for $k \geq 0$ such that $h' \neq h$ and $h - h' = \delta > 0$. Since $y_{\alpha,k}(n, s)$ is non-decreasing in n , it converges asymptotically to $\frac{h'}{\alpha}$ as $n \rightarrow \infty$. For $\frac{\delta}{\nu} > 0$, there exist a finite n'_s for each s such that for all $n \geq n'_s$, we have $y_{\alpha,k}(n, s) > \frac{h'}{\alpha} - \frac{\delta}{\nu}$. Replacing $y_{\alpha,k}(n, s)$ by $\frac{h'}{\alpha} - \frac{\delta}{\nu}$, replacing the optimal μ_{n-1}^* by the suboptimal μ_n^* , and replacing the optimal $u_{i,n-1}$ by the suboptimal $u_{i,n}$ in the Equation 21, we obtain:

$$\begin{aligned}
& y_{\alpha,k+1}(n, s) \\
& > h + \sum_{i=1}^l \lambda_{i,s} \left\{ (1 - u_{i,n})\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) + u_{i,n}\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) \right\} + \sum_{i=1}^l (\lambda_{max} - \lambda_{i,s})\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) \\
& + \sum_{s' \neq s \in \mathbb{S}} Q_{ss'}\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) + (\eta + Q_{ss})\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) + \mu_n^*\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) + (\mu_{max} - \mu_n^*)\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) \\
& = h + (l\lambda_{max} + \eta + \mu_{max})\left(\frac{h'}{\alpha} - \frac{\delta}{\nu}\right) \\
& = \left(\frac{h'}{\alpha}\right) \quad \text{for all } n > n'_s \text{ and } s \in \mathbb{S}
\end{aligned}$$

which is a contradiction since we assumed $\frac{h'}{\alpha}$ is the least upper bound of $y_{\alpha,k+1}(n, s)$. Thus, $\frac{h}{\alpha}$ is the least upper bound of $y_{\alpha,k+1}(n, s)$ for all $n \in \mathbb{Z}^+$ and $s \in \mathbb{S}$. Finally, using Lemma 1, we extend the result to the infinite horizon case, i.e. $\frac{h}{\alpha}$ is the least upper bound of $y_{\alpha}(n, s)$.

Similarly, we can show that $-p_1$ is a strict lower bound of $y_\alpha(n, s)$ for all $n \in \mathbb{Z}^+$ and $s \in \mathbb{S}$. $\square$

E Proof of Theorem 1

Theorem. The infinite horizon discounted cost optimal problem (Equation 6) has a structured optimal policy characterized by a season dependent inventory threshold vector $\mathbf{z}_s = [z_{1,s}, \dots, z_{l,s}, z_{\mu,s}, z_{b,s}]$ such that for each season s , it is optimal to:

1. Satisfy an arriving class i demand only if the inventory level is above $z_{i,s}$ for $i \in \{1, \dots, l\}$
2. Produce at rate $\mu_{\max}$ if the inventory level drops below $z_{\mu,s}$
3. Produce at rate μ_{reg} if the inventory level drops below $z_{b,s}$ but is greater than or equal to $z_{\mu,s}$
4. Stop production if the inventory level equals or exceeds $z_{b,s}$

Proof. The right side of Equation 6 depends on both the inventory allocation decision $u_i(n, s)$ and the production rate control decision $\mu(n, s)$. Since the inventory allocation and the production control decisions are independent of each other, we minimize the $u_i(n, s)$ dependent terms and $\mu(n, s)$ terms separately in order to minimize the right side of Equation 6.

To determine the optimal inventory allocation, consider the demand arrival terms of customers of any class i in right side of Equation 6, i.e,

$$\min_{u_i \in \{0,1\}} \left\{ -u_i(n, s)(v_\alpha(n, s) + p_i) + u_i(n, s)H_i v_\alpha(n, s) \right\} \quad (24)$$

Note, $u_i(0, s) = 0$. When $n > 0$, we obtain

$$\begin{aligned} & \min_{u_i \in \{0,1\}} \left\{ -u_i(n, s)(v_\alpha(n, s) + p_i) + u_i(n, s)H_i v_\alpha(n, s) \right\} \\ &= \min_{u_i \in \{0,1\}} \left\{ -u_i(n, s)(v_\alpha(n, s) + p_i) + u_i(n, s)v_\alpha(n - 1, s) \right\} \\ &= \min_{u_i \in \{0,1\}} -u_i(n, s) \left\{ y_\alpha(n, s) + p_i \right\} \end{aligned} \quad (25)$$

From Proposition 1, $y_\alpha(n, s) + p_i$ is a non-decreasing function of n for a given s . Let $z_{i,s}$

denote the minimum n at which

$$y_\alpha(n, s) + p_i > 0 \text{ for all } n > z_{i,s} \quad (26)$$

From Equations 25 and 26, we conclude that it is optimal to satisfy an arriving class i demand, i.e., $u_i(n, s) = 1$, only if $n > z_{i,s}$, and to reject the demand, i.e., $u_i(n, s) = 0$, when $n \leq z_{i,s}$.

To determine the optimal production control, we choose only the μ dependent terms from right side of Equation 6 and apply the first order minimization condition to obtain the expression for the optimal policy $\mu^*(n, s)$

$$\mu^*(n, s) = \arg \min_{\mu \in [0, \mu_{max}]} \left(c(\mu) + \mu y_\alpha(n+1, s) \right) \quad (27)$$

From Equation 27 and by exploiting the piece-wise linearity of the production cost function (Equation 2), we obtain the optimal production rate as a function of $y_\alpha(n+1, s)$: $\mu^*(n, s) = 0$ if $-c_{reg} < y_\alpha(n+1, s)$; $\mu^*(n, s) = \mu_{reg}$ if $-c_{max} < y_\alpha(n+1, s) \leq -c_{reg}$; and $\mu^*(n, s) = \mu_{max}$ if $y_\alpha(n+1, s) \leq -c_{max}$.

By taking inverse of $y_\alpha(\cdot, s)$, we obtain the optimal production rate for the entire range of the inventory n for a given s . For ease of notation, we replace $y_\alpha(n+1, s)$ with y in the rest of this proof. Since the function y is not strictly monotone, its inverse may have multiple entries. To satisfy the above inequalities, we choose the maximum of these entries. Denote $\lfloor \max y^{-1}(-c_{reg}) \rfloor$ as $z_{b,s}$ and $\lfloor \max y^{-1}(-c_{max}) \rfloor$ as $z_{\mu,s}$. Expressed in terms of $z_{b,s}$ and $z_{\mu,s}$, it is optimal to stop production when $z_{b,s} \leq n$, to produce at rate μ_{reg} when $z_{\mu,s} \leq n < z_{b,s}$, and to produce at rate μ_{max} when $n < z_{\mu,s}$.

Since $y_\alpha(n, s)$ is non-decreasing in n for a fixed s , we also conclude that $z_{b,s}$ and $z_{\mu,s}$ are the respective least values of n such that

$$\begin{aligned} y_\alpha(n, s) + c_{reg} &> 0 \text{ for all } n > z_{b,s} \\ y_\alpha(n, s) + c_{max} &> 0 \text{ for all } n > z_{\mu,s} \end{aligned} \quad (28)$$

□